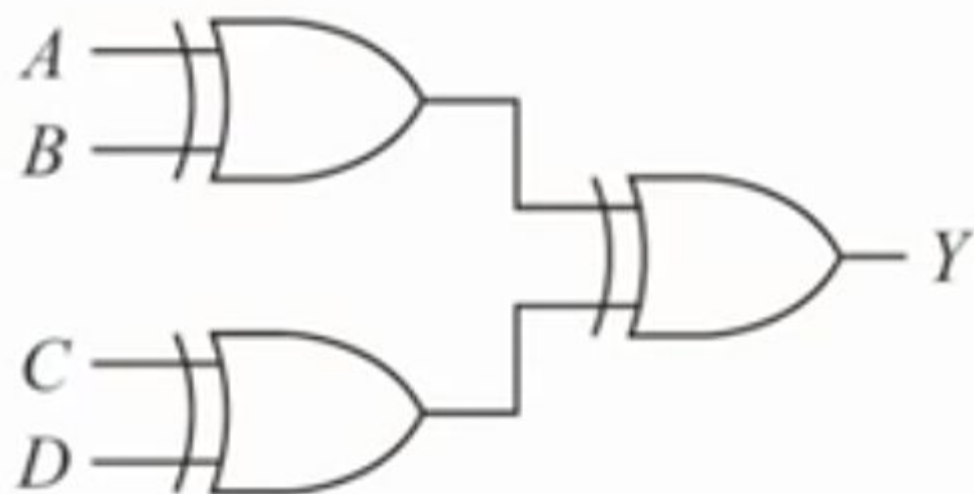


Error Detection and Correction Codes

<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>Y</i>
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0



- **Even Parity:** In the input bits, there is even number of 1s.

Example: For, 4-bit input: 1001, 1100, 1111

8-bit input: 11000101, 01111011

16-bit input: 1011001110110011

- **Odd Parity:** In the input bits, there is odd number of 1s.

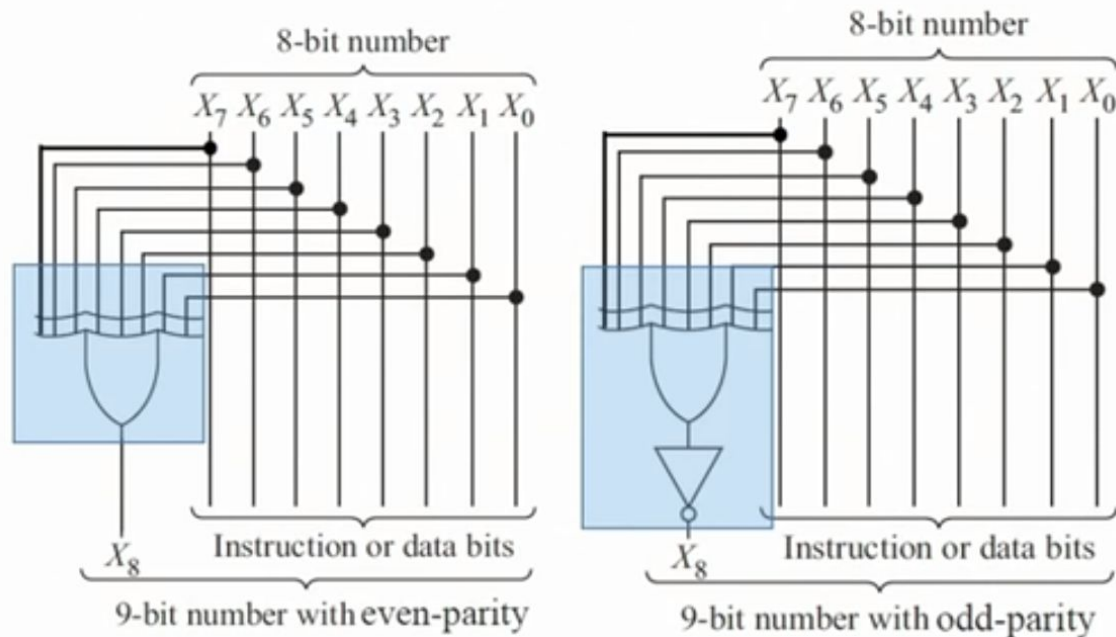
Example: For, 4-bit input: 0001, 1101, 1110

8-bit input: 11000001, 01011011

16-bit input: 1011001010110011

- Prior information about parity (even or odd) in the message can help in detecting **1-bit error**.

Parity Generation



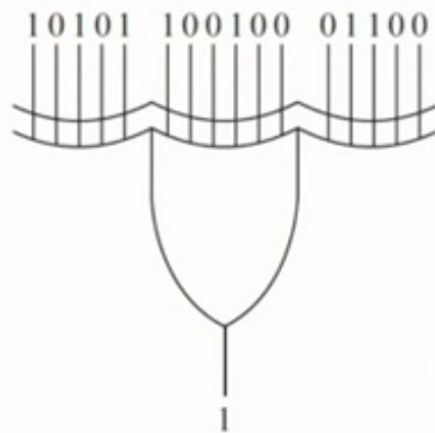
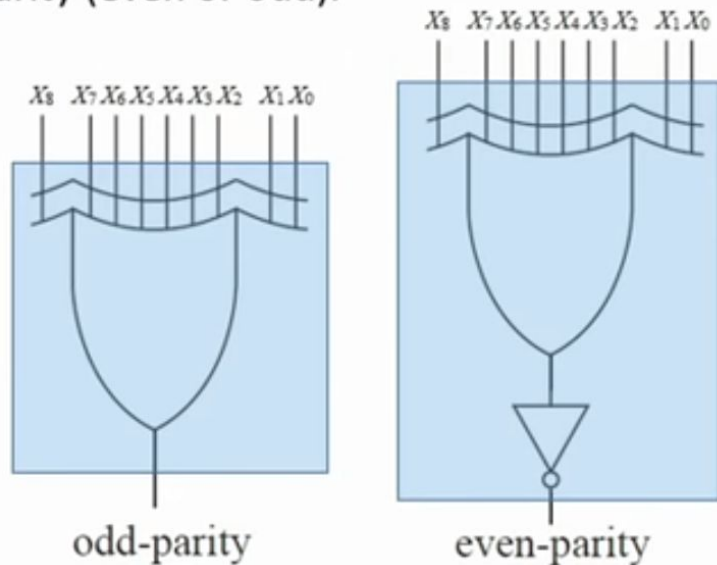
- The aim here is to add a parity bit with the message so that message and parity bit together make it of a predefined parity (even or odd).
- The message is random i.e. can be of even or odd parity.

Example: Instead of 8 bits (message), 9 bits (message + parity bit) sent.

Decimal	8421 BCD				Odd parity	Even parity
0	0	0	0	0	1	0
1	0	0	0	1	0	1
2	0	0	1	0	0	1
3	0	0	1	1	1	0
4	0	1	0	0	0	1
5	0	1	0	1	1	0
6	0	1	1	0	1	0
7	0	1	1	1	0	1
8	1	0	0	0	0	1
9	1	0	0	1	1	0

Parity Checking

For 9-bit data received with pre-defined parity (even or odd).



Example of 16-bit odd parity checking

Odd parity checking by
Ex-OR, Odd parity
generation by Ex-NOR
(i.e. EX-OR then NOT)

Similarly, for Even Parity.

Hamming Distance and Parity Code

- The number of bit positions by which code words differ is called Hamming distance.
- It can be found by counting number of 1s in bit-wise Ex-OR of two code words.

Parity bit
↓

Data 1: 10100011	Code word 1: 010100011	} Even parity coded
Data 2: 11011001	Code word 2: 111011001	

Bit-wise Ex-Or output: 101111010 : Hamming distance, $d = 6$

Data 1: 00000000	Code word 1: 000000000
Data 2: 00000001	Code word 2: 100000001

Bit-wise Ex-Or output: 100000001 : Hamming distance, $d = 2$

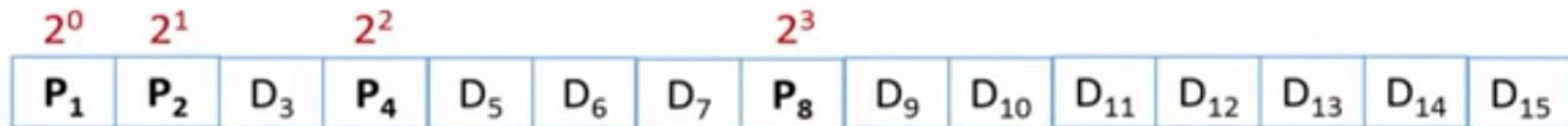
Parity Code:
Data with a parity bit

↓
Can detect 1 bit error.
Cannot correct any.

- To detect b bit errors, minimum hamming distance, $d_{min} = b + 1$.
- To correct b bit errors, minimum hamming distance, $d_{min} = 2b + 1$.

Hamming Code

- Useful for correcting 1-bit error. It uses more than 1 parity bits.
- Parity bits in the code word are positioned at 2^i -th positions.
Rest of the positions are filled by data bits.



If m parity bits ($m \geq 2$) are used
to code n data bits then,

$$2^m > m + n$$

Hamming Code Generation

P ₁	P ₂	D ₃	P ₄	D ₅	D ₆	D ₇
----------------	----------------	----------------	----------------	----------------	----------------	----------------

Binary coded position 0001 0010 0011 0100 0101 0110 0111

(7,4) Code: Rate = 4/7

$$P_1 = D_3 \oplus D_5 \oplus D_7 \quad : \text{All with 1 in 1's place of position}$$

$$P_2 = D_3 \oplus D_6 \oplus D_7 \quad : \text{All with 1 in 2's place of position}$$

$$P_4 = D_5 \oplus D_6 \oplus D_7 \quad : \text{All with 1 in 4's place of position}$$

Extending:

P ₁	P ₂	D ₃	P ₄	D ₅	D ₆	D ₇	P ₈	D ₉	D ₁₀	D ₁₁	D ₁₂
----------------	----------------	----------------	----------------	----------------	----------------	----------------	----------------	----------------	-----------------	-----------------	-----------------

0001 0010 0011 0100 0101 0110 0111 1000 1001 1010 1011 1100

$$P_1 = D_3 \oplus D_5 \oplus D_7 \oplus D_9 \oplus D_{11} \quad \dots$$

$$P_2 = D_3 \oplus D_6 \oplus D_7 \oplus D_{10} \oplus D_{11} \quad \dots$$

$$P_4 = D_5 \oplus D_6 \oplus D_7 \oplus D_{12} \quad \dots$$

$$P_8 = D_9 \oplus D_{10} \oplus D_{11} \oplus D_{12} \quad : \text{All with 1 in 8's place of position}$$

$$d_{min} = 3$$

Can detect all 2 bit error.

Can correct all 1 bit error.

Data: 10110101

P₁	P₂	1	P₄	0	1	1	P₈	0	1	0	1
0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100

$$P_1 = D_3 \oplus D_5 \oplus D_7 \oplus D_9 \oplus D_{11} = 1 \oplus 0 \oplus 1 \oplus 0 \oplus 0 = 0$$

$$P_2 = D_3 \oplus D_6 \oplus D_7 \oplus D_{10} \oplus D_{11} = 1 \oplus 1 \oplus 1 \oplus 1 \oplus 0 = 0$$

$$P_4 = D_5 \oplus D_6 \oplus D_7 \oplus D_{12} = 0 \oplus 1 \oplus 1 \oplus 1 = 1$$

$$P_8 = D_9 \oplus D_{10} \oplus D_{11} \oplus D_{12} = 0 \oplus 1 \oplus 0 \oplus 1 = 0$$

Coded Data: 001101100101

One bit Error Correction

$$C_1 = D_3 \oplus D_5 \oplus D_7 \oplus D_9 \oplus D_{11} \oplus P_1$$

$$C_2 = D_3 \oplus D_6 \oplus D_7 \oplus D_{10} \oplus D_{11} \oplus P_2$$

$$C_4 = D_5 \oplus D_6 \oplus D_7 \oplus D_{12} \oplus P_4$$

$$C_8 = D_9 \oplus D_{10} \oplus D_{11} \oplus D_{12} \oplus P_8$$

- If $C_8C_4C_2C_1 = 0000$: No error
- Else, it gives position of the error bit
e.g. $C_8C_4C_2C_1 = 0111$ means 7th bit is erroneously received
- Invert the erroneous bit to correct.

Example:

Coded Data: 001101100101

Data with 1-bit error: 001101100111

In 11th pos.

$$C_1 = 1 \oplus 0 \oplus 1 \oplus 0 \oplus 1 \oplus 0 = 1$$

$$C_2 = 1 \oplus 1 \oplus 1 \oplus 1 \oplus 1 \oplus 0 = 1$$

$$C_4 = 0 \oplus 1 \oplus 1 \oplus 1 \oplus 1 = 0$$

$$C_8 = 0 \oplus 1 \oplus 1 \oplus 1 \oplus 0 = 1$$

1011: 11th bit 1 is erroneous
It is to be made 0 to correct

Note: If $C_8C_4C_2C_1 = 1000$ then 8th position bit i.e. P_8 is erroneous.

Two bit Error Detection

$$C_1 = D_3 \oplus D_5 \oplus D_7 \oplus D_9 \oplus D_{11} \oplus P_1$$

$$C_2 = D_3 \oplus D_6 \oplus D_7 \oplus D_{10} \oplus D_{11} \oplus P_2$$

$$C_4 = D_5 \oplus D_6 \oplus D_7 \oplus D_{12} \oplus P_4$$

$$C_8 = D_9 \oplus D_{10} \oplus D_{11} \oplus D_{12} \oplus P_8$$

$$C_8 C_4 C_2 C_1 = 0010$$

$$C_1 = D_3 \oplus D_5 \oplus D_7 \oplus D_9 \oplus D_{11} \oplus P_1$$

$$C_2 = D_3 \oplus D_6 \oplus D_7 \oplus D_{10} \oplus D_{11} \oplus P_2$$

$$C_4 = D_5 \oplus D_6 \oplus D_7 \oplus D_{12} \oplus P_4$$

$$C_8 = D_9 \oplus D_{10} \oplus D_{11} \oplus D_{12} \oplus P_8$$

$$C_8 C_4 C_2 C_1 = 0110$$

← Two bits flip

$$C_1 = D_3 \oplus D_5 \oplus D_7 \oplus D_9 \oplus D_{11} \oplus P_1$$

$$C_2 = D_3 \oplus D_6 \oplus D_7 \oplus D_{10} \oplus D_{11} \oplus P_2$$

$$C_4 = D_5 \oplus D_6 \oplus D_7 \oplus D_{12} \oplus P_4$$

$$C_8 = D_9 \oplus D_{10} \oplus D_{11} \oplus D_{12} \oplus P_8$$

$$C_8 C_4 C_2 C_1 = 0001$$

$$C_1 = D_3 \oplus D_5 \oplus D_7 \oplus D_9 \oplus D_{11} \oplus P_1$$

$$C_2 = D_3 \oplus D_6 \oplus D_7 \oplus D_{10} \oplus D_{11} \oplus P_2$$

$$C_4 = D_5 \oplus D_6 \oplus D_7 \oplus D_{12} \oplus P_4$$

$$C_8 = D_9 \oplus D_{10} \oplus D_{11} \oplus D_{12} \oplus P_8$$

$$C_8 C_4 C_2 C_1 = 0000$$

← Three bits flip